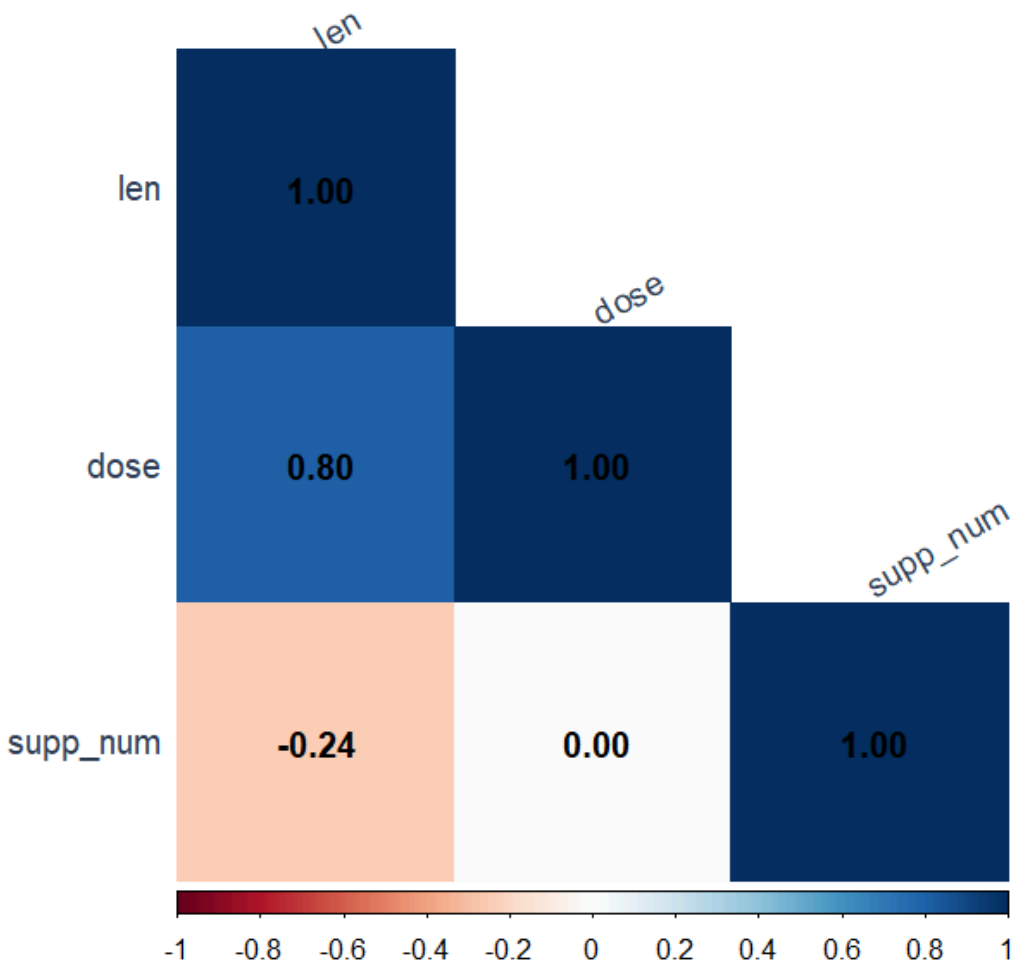
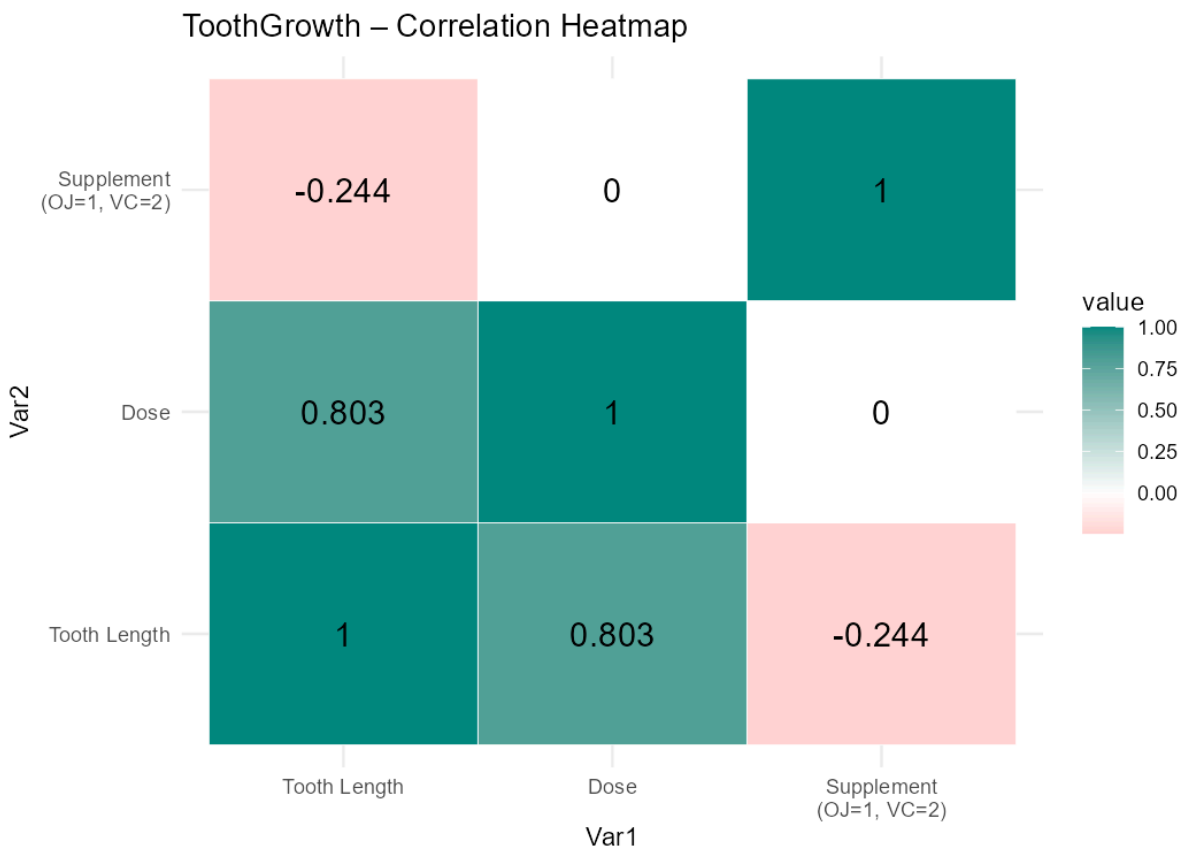
Lab 9

ToothGrowth





- **Strong Positive Correlation (0.80):** There is a strong positive linear relationship between `dose` and tooth length (`len`). As the vitamin C dose increases, tooth length significantly increases.
- **Weak Negative Correlation (-0.24):** There is a weak negative relationship between the supplement type (`supp_num`) and tooth length, indicating that the delivery method has a minor impact compared to the dosage.
- **Zero Correlation (0.00):** There is absolutely no correlation between `dose` and supplement type, which is expected since they are independent experimental variables.
- **Scatter Plot Insight:** The scatter plot visually confirms the strong positive correlation, showing an upward trend for both supplement types (OJ and VC) as the dosage increases.

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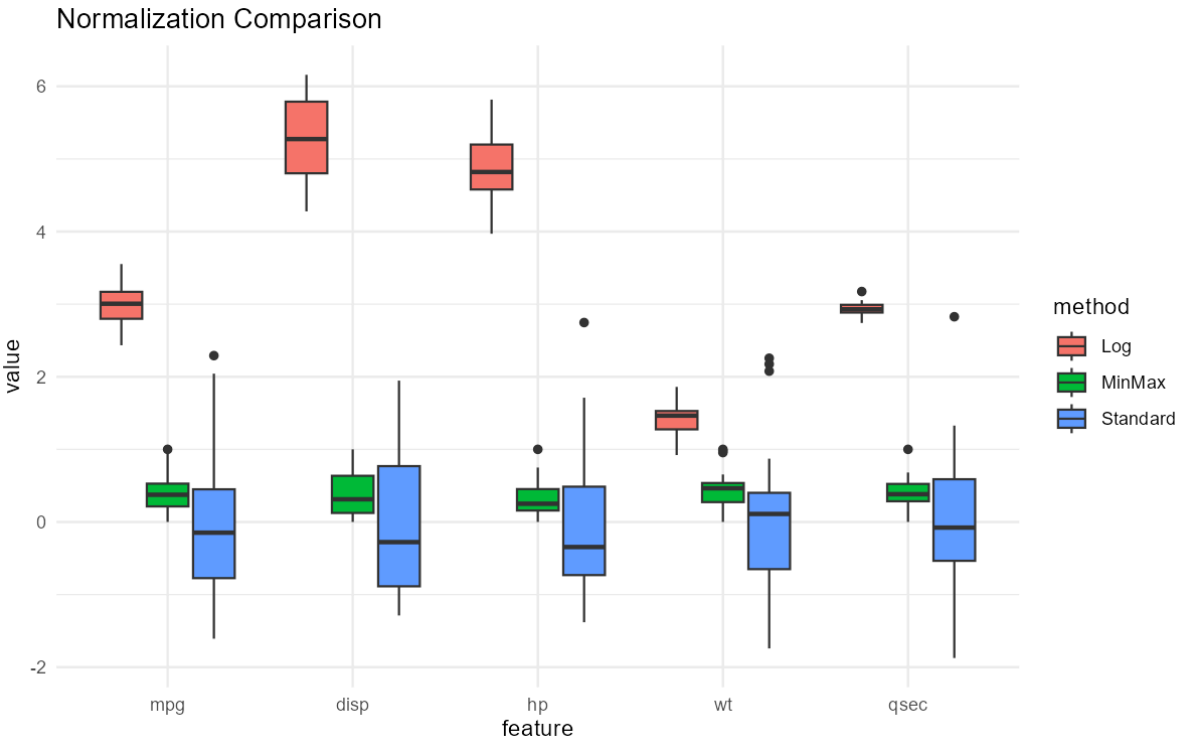
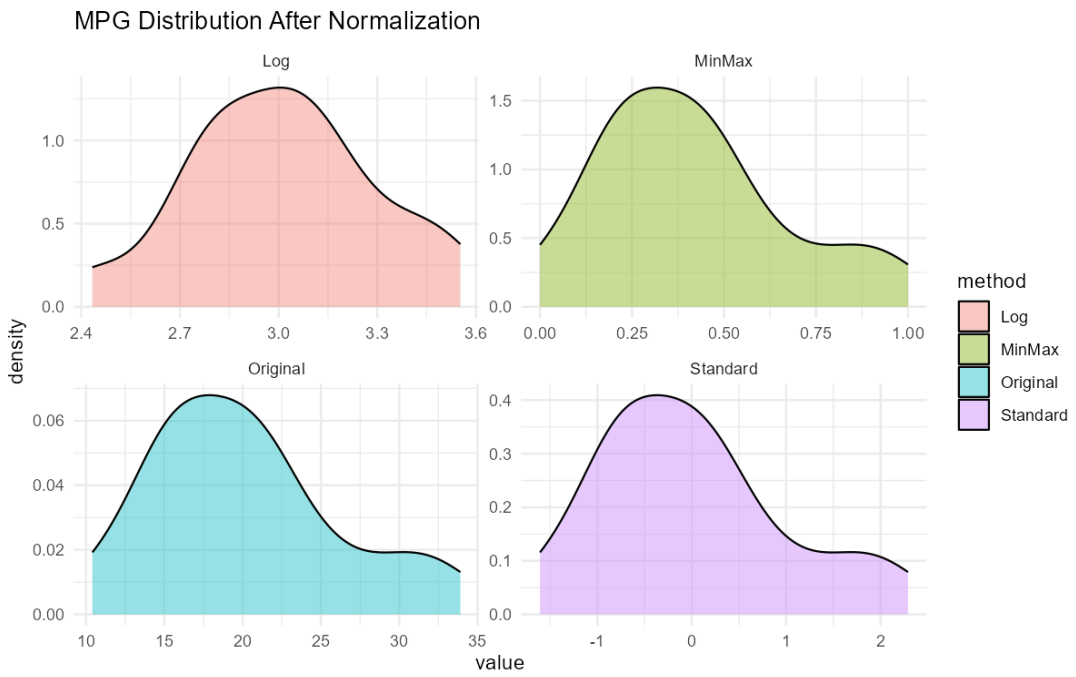
mtcars



- **The Need for Normalization:** The original dataset contains features with vastly different scales (e.g., `wt` is around 1-5, while `disp` goes up to 472). Normalization makes these features directly comparable without large values dominating the model.
- **Log Transformation:** * Compresses wide ranges of data.
    - As seen in the density plots, it helps reduce positive skewness, making the distribution of variables like `mpg` more bell-shaped/symmetric.
- **Min-Max Scaling:**
    - Strictly bounds all data points between exactly **0 and 1**.
    - It perfectly preserves the original shape of the data's distribution (the density plot shape is identical to the original, just shifted to a 0-1 scale).



- **Standard Scaling (Z-score):**
  - Centers the data around a **mean of 0** and a **standard deviation of 1**.
  - Unlike Min-Max, it does not confine data to a specific boundary (values range roughly from -2 to +3), making it more robust to outliers.